

Finite difference method

In a certain interest rate model, the price of a risk-free bond with maturity date $T > 0$ is given at time $t \in [0, T]$ by the function $F(t, r)$, where r is the value of the short-term interest rate. Under the no-arbitrage condition it satisfies

$$F(T, r) = 1$$

for all $r \geq 0$, together with the partial differential equation

$$\frac{\partial F}{\partial t}(t, r) + a(b - r)\frac{\partial F}{\partial r}(t, r) + \frac{1}{2}\sigma^2 r \frac{\partial^2 F}{\partial r^2}(t, r) - rF(t, r) = 0$$

for all $t \in (0, T)$, $r \geq 0$. Here a , b and σ are given parameters.

The following facts are generally known about the function F :

- Theoretical bond prices are usually between 0 and 1, in other words, it is safe to assume that $F(t, r) \in [0, 1]$ for all $t \in [0, T]$, $r \geq 0$.
- Interest rates are usually small, in other words, it is safe to assume that $r \leq 0.5 = 50\%$ in practical applications.
- The behaviour of F when interest rates are small can be modelled as

$$\lim_{r \rightarrow 0} F(t, r) = A(t)$$

for all $t \in [0, T)$ where

$$A(t) = \left[\frac{2\gamma e^{(a+\gamma)t/2}}{(\gamma + a)(e^{\gamma t} - 1) + 2\gamma} \right]^{2ab/\sigma^2}$$

and $\gamma = \sqrt{a^2 + 2\sigma^2}$.

- The behaviour of $\frac{\partial F}{\partial r}(t, r)$ when rates are large is asymptotically similar to $-A(t)B(t)e^{-B(t)r}$, where

$$B(t) = \frac{2(e^{\gamma t} - 1)}{(\gamma + a)(e^{\gamma t} - 1) + 2\gamma}$$

and A and γ are as before. In mathematical terms this can be expressed as

$$\lim_{r \rightarrow \infty} \left(\frac{\partial F}{\partial r}(t, r) + A(t)B(t)e^{-B(t)r} \right) = 0$$

Exercise 3.1 (Backward Euler method: 30% of project mark). Design a backward Euler finite difference scheme to approximate the solution to the partial differential equation in Equation 1 above. Your work should include the following:

1. Convert the final value problem into an initial value problem.

Let $s = T - t$ then $F(T - s, r) = G(s, r)$ such that $G(0, r) = 1$. Our PDE becomes

$$\partial_s G - a(b - r)\partial_r G - \frac{1}{2}\sigma^2 r \partial_r^2 G + rG = 0 \tag{1}$$

with the following properties of G :

$$\lim_{r \rightarrow 0} G(s, r) = A(s) \qquad \lim_{r \rightarrow \infty} \left(\partial_r G + A(s)B(s)e^{-B(s)r} \right) = 0$$

where $A(s)$ and $B(s)$ are as previously defined.

2. Propose appropriate boundary conditions for r . You should take the minimum value of r to be 0, and the maximum value to be r_{max} (to be chosen later). Similarly to before, we have the boundary at $r = 0$

$$G(s, 0) = A(s)$$

Using the limit as $r \rightarrow \infty$, we can estimate the limit by writing

$$\lim_{r \rightarrow \infty} \partial_r G = -A(s)B(s)e^{-B(s)r}$$

and by integrating in r we get our upper boundary condition at $r = r_{max}$

$$G(s, r_{max}) = A(s)e^{-B(s)r_{max}}$$

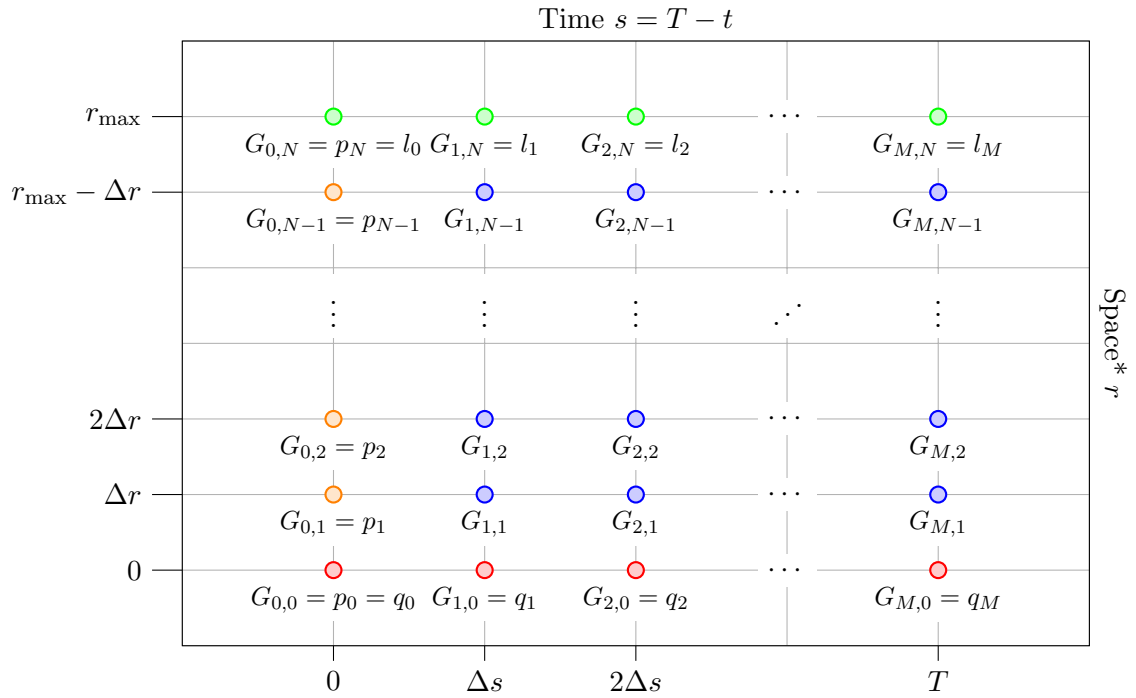
3. Define a suitable grid to discretize the domain of F , and derive an iterative scheme in matrix form that would allow you to approximate the value of F at the grid points, using the boundary values.

Define $\Delta s = T/M$ and $\Delta r = r_{max}/N$. Then $s_i = i\Delta s$ and $r_j = j\Delta r$. The discretised domain is then:

$$\bar{D} = \{(s_i, r_j) : i = 0, \dots, M, j = 0, \dots, N\} \text{ for some } M, N \in \mathbb{N}$$

so

$$G(s_i, r_j) = G_{i,j} \quad p_j = p(r_j) \quad q_i = q(t_i) \quad l_i = l(t_i)$$



Now we want to derive the iterative scheme for this problem. First we must write the backwards difference approximation for the time derivative (note that the time derivative is now referring to ∂_s after our substitution) and the central difference approximation for the spacial derivatives. Let $G(s_i, r_j) = G_{i,j}$ and $\frac{\partial}{\partial a} \equiv \partial_a$.

$$\partial_s G(s_{i+1}, r_j) = \frac{1}{\Delta s} (G_{i+1,j} - G_{i,j}) + O(\Delta s) \quad (2)$$

$$\partial_r G(s_{i+1}, r_j) = \frac{1}{2\Delta r} (G_{i+1,j+1} - G_{i+1,j-1}) + O(\Delta r^2) \quad (3)$$

$$\partial_r^2 G(s_{i+1}, r_j) = \frac{1}{(\Delta r)^2} (G_{i+1,j+1} - 2G_{i+1,j} + G_{i+1,j-1}) + O((\Delta r)^2) \quad (4)$$

Now we take Equations 2, 3 and 4 and substitute them back into the PDE for G Equation 1, (discarding higher order terms) and simplify. The following steps are such simplification.

$$\frac{(G_{i+1,j} - G_{i,j})}{\Delta s} - \frac{a(b-r)(G_{i+1,j+1} - G_{i+1,j-1})}{2\Delta r} - \frac{\sigma^2 r (G_{i+1,j+1} - 2G_{i+1,j} + G_{i+1,j-1})}{2(\Delta r)^2} + rG_{i+1,j} = 0$$

Expanding

$$\frac{G_{i+1,j}}{\Delta s} - \frac{G_{i,j}}{\Delta s} - \frac{a(b-r)G_{i+1,j+1}}{2\Delta r} + \frac{a(b-r)G_{i+1,j-1}}{2\Delta r} - \frac{\sigma^2 r G_{i+1,j+1}}{2(\Delta r)^2} + 2\frac{\sigma^2 r G_{i+1,j}}{2(\Delta r)^2} - \frac{\sigma^2 r G_{i+1,j-1}}{2(\Delta r)^2} + rG_{i+1,j} = 0$$

Isolating $G_{i,j}$ term and collecting like-terms simplifies into the following

$$\begin{aligned} \frac{G_{i,j}}{\Delta s} = & \left[-\frac{a(b-r)}{2\Delta r} - \frac{\sigma^2 r}{2(\Delta r)^2} \right] G_{i+1,j+1} + \\ & + \left[\frac{1}{\Delta s} + \frac{2\sigma^2 r}{2(\Delta r)^2} + r \right] G_{i+1,j} + \\ & + \left[\frac{a(b-r)}{2\Delta r} - \frac{\sigma^2 r}{2(\Delta r)^2} \right] G_{i+1,j-1} \end{aligned}$$

So we can write these as functions of r as our given problem keeps a , b and σ constant

$$\begin{aligned} G_{i,j} = & \underbrace{\left[-\frac{a(b-r)\Delta s}{2\Delta r} - \frac{\sigma^2 r \Delta s}{2(\Delta r)^2} \right]}_{\delta(r)} G_{i+1,j+1} + \\ & + \underbrace{\left[1 + \frac{2\sigma^2 r \Delta s}{2(\Delta r)^2} + r \Delta s \right]}_{\alpha(r)} G_{i+1,j} + \\ & + \underbrace{\left[\frac{a(b-r)\Delta s}{2\Delta r} - \frac{\sigma^2 r \Delta s}{2(\Delta r)^2} \right]}_{\beta(r)} G_{i+1,j-1} \end{aligned}$$

and finally

$$G_{i,j} = \delta(r_j) G_{i+1,j+1} + \alpha(r_j) G_{i+1,j} + \beta(r_j) G_{i+1,j-1} \quad (5)$$

where

$$\begin{aligned} \delta_j = \delta(r_j) &= -\frac{\Delta s}{2\Delta r} \left(a(b-r_j) + \frac{\sigma^2 r_j}{\Delta r} \right), & \alpha_j = \alpha(r) &= 1 + \sigma^2 r_j \frac{\Delta s}{(\Delta r)^2} + r_j \Delta s, \\ \beta_j = \beta(r_j) &= -\frac{\Delta s}{2\Delta r} \left(\frac{\sigma^2 r_j}{\Delta r} - a(b-r_j) \right) \end{aligned}$$

we can now use these functions to create the iterative step. The algorithm is as follows:

- Write $\mathbf{G}_i = [G_{i,1}, \dots, G_{i,N-1}]^T$ for each time step $i = 0, \dots, M$
- The initial value $\mathbf{G}_0$ is determined by the initial condition: $G_{0,j} = [1, \dots, 1]^T$

- Recall that $G_{i,0}$ and $G_{i,N}$ are determined by the initial and boundary conditions $\forall i: G_{i,0} = A(s_i)$ and $G_{i,N} = A(s_i)e^{-B(s_i)r_{\max}}$

The iterative step is as follows: Equation 5 can be written in vector form as

$$\begin{bmatrix} \alpha_1 & \delta_1 & 0 & \dots & 0 \\ \beta_2 & \alpha_2 & \delta_2 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & \beta_{N-2} & \alpha_{N-2} & \delta_{N-2} \\ 0 & \dots & 0 & \beta_{N-1} & \alpha_{N-1} \end{bmatrix} \begin{bmatrix} G_{i+1,1} \\ \vdots \\ G_{i+1,N-1} \end{bmatrix} + \begin{bmatrix} \beta_1 G_{i+1,0} \\ 0 \\ \vdots \\ 0 \\ \delta_{N-1} G_{i+1,N} \end{bmatrix} = \begin{bmatrix} G_{i,1} \\ \vdots \\ G_{i,N-1} \end{bmatrix} \quad (6)$$

In other words,

$$A_{\text{BE}} \mathbf{G}_{i+1} - \mathbf{a}_{i+1} = \mathbf{G}_i \quad (7)$$

where:

- A_{BE} is an $(N-1) \times (N-1)$ tridiagonal matrix.
- $\mathbf{a}_{i+1}$ is an $(N-1)$ -dimensional vector that depends on $G_{i+1,0}$ and $G_{i+1,N}$.

Finally, the Backward Euler algorithm for solving our new PDE in Equation 1:

- Choose M and N and calculate $\Delta s = T/M$ and $\Delta r = r_{\max}/N$
- Set $\mathbf{G}_0 = [1, \dots, 1]^T$ according to our initial condition
- Set $G_{0,0}$ and $G_{0,N}$ according to our initial and boundary conditions for r :

$$G_{0,0} = A(s_0) = 1, \quad G_{0,N} = A(s_0)e^{-B(s_0)r_{\max}} = 1.$$

- For $i = 0, \dots, M-1$, do the following:
 - Set $G_{i+1,0} = A(s_{i+1})$ and $G_{i+1,N} = A(s_{i+1})e^{-B(s_{i+1})r_{\max}}$
 - Solve Equation 7 to determine $\mathbf{G}_{i+1}$. As A_{BE} is invertible, we have

$$\mathbf{G}_{i+1} = A_{\text{BE}}^{-1}(\mathbf{G}_i + \mathbf{a}_{i+1})$$

We note that this method is stable if the absolute value of the eigenvalues of A_{BE}^{-1} are less than 1.

- Using the parameter values $a = 2$, $b = 0.05$, $\sigma = 0.2$ and $T = 1.5$, use your function to approximate $F(0, r)$ for $r \in (0, 0.15)$ and $t \in [0, T]$ and display the results graphically. Choose M and N such that the iterative scheme is stable, and choose an appropriate value of $r_{\max}$. Give details of your reasoning. (see ‘part.3.ipynb’ for displayed results).

The reason we set $r_{\max} = 0.5$ is guided by the question. As interest rates are usually small, we are told that it is safe to assume $r \leq 0.5$ in practice. $r_{\max}$ should be large enough that the approximated upper boundary condition does not significantly affect the approximations for $r \in (0, 0.15)$, while also staying within a range meaningful to the problem.

As the Backwards Euler scheme is unconditionally stable, any choice of M and N result in a stable approximation. Therefore I chose M and N so that they are large enough to produce a smooth approximation, but not so large that they have a significant computational cost. Additionally, I choose to set $M = 50$ and $N = 3M = 150$ to ensure a smooth resolution in the r -direction as we only consider a slice in r of the total surface.